MIL-L-48765 (AR)
Amendment 2
17 December 1979
Superseding
Amendment 1
25 September 78

MILITARY SPECIFICATION

LITHIUM HEXAFLUORORSENATE

This Amendment forms a part of Military Specification MIL-L-48765(AR) dated 19 June 1975.

Page 1

Delete "(PA)" wherever it appears and substitute "(AR)"

Delete the Preamble and substitute the following:

"This specification approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense".

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Para. 3.1.2. X-ray diffraction. Delete Table I in its entirety and substitute the following:

"Table I

Peak, d-value (decreasing order of intensity)

3.30 - 3.70
4.40 - 4.65
2.90 - 3.10
1.65 - 1.76"

Para. 3.1.3 Infra-red spectroscopy. Add the following" "This requirement may be used as an alternate to x-ray diffraction requirement given in 3.1.2."

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Para 3.1.4 Impurities - Delete the last sentence in paragraph.

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Delete Table III in its entirety and substitute the following Table III:

Table III

Impurities	Maximum Allowable Concentration, PPM	Test Paragraph
Total Halogens	50	4.4.3.6
Water	150	4.4.3.10"

Para. 3.1.5 Properties. Delete Hydrolysis/Dehydration and Particle Size Requirement in their entirety and change "Potential" to "Potentiometric Value".

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Para. 4.4.2 Testing. At the end of the sentence add the following: "At a Relative Humidity not to exceed 4% at ambient conditions:.

Para. 4.4.2.1 Sampling. The following changes shall be made:
1st sentence. Delete "1 pound" and substitute "100 grams".
2nd sentence. At the end of the sentence add the following: "or the samples from each batch that is combined into a shipping lot shall be subjected to the tests listed below in accordance with MIL-STD-105. Level II."
Table - Delete Table in its entirety and substitute the following Table V.

"Table V

Inspection/Requirement	Defect Classification
Form & Color (see 3.1.1)	Major
X-ray Diffraction (see 3.1.2) or	Major
Infra-red Spectroscopy (see 3.1.3)	Major
(Alternate Method)	
Total Halogens (see 3.1.4)	Major
Water (see 3.1.4)	Major
pH (see 3.1.5)	Major
Solubility in MF (see 3.1.5)	Major
Potentiometric Value (see 3.1.5)	Major"

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Para 4.4.3.1 Form. Delete the first sentence and substitute the following sentence:

"Spread the sample on a flat surface and examine the material microscopically (at a 25X magnification) for the presence of contamination and form.

Paragraphs 4.4.3.4, 4.4.3.5, 4.4.3.7, 4.4.3.8, 4.4.3.9, 4.4.3.13 and 4.4.3.14 - Delete these paragraphs in their entirety.

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Para. 4.4.3.11 pH. Make the following changes to this paragraph:

1st Sentence: Prepare a 1 Molar.....given in ASTM-E70" delete in its entirety and substitute: "Prepare a 1 Molar solution of the $LiAsf_6$ by dissolving 195.86g of the sample in a liter of boiled distilled water (pH 6-7) and determine the pH of the resultant solution in accordance with the method given in ASTM-E70. The pH measurement should be taken immediately following preparation of the solution, with reading taken between 1 and 3 minutes following preparation.

3rd Sentence: Delete "distilled water" and substitute: "boiled distilled water (pH 6-7)."

4th Sentence: At end of sentence add "the pH measurement shall be taken immediately following preparation of the solution,with readings taken between 1 and 3 minutes following preparation.

5th Sentence: "The pH measurement.....for a least 10 minutes period.." delete this sentence in its entirety.

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Para 4.4.3.14 Particle Size. Delete this paragraph in its entirety.

Para 4.4.3.15 Potentiometric Value. Substitute the following method:

"4.4.3.6 Halogens: Delete in its entirety and substitute the following:

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"4.4.3.6 Total Halogens. Determine the halogens present in the compound using a spectrophotometric described in the literature or reference textbook (see 6.4).

"4.4.3.15 Potentiometric Value

The potentiometric value (EMF) shall be checked on a 2 molar solution of the sample in methyl formate (see 4.4.3.11).

4.4.3.15.1 Equipment

a) Platinum electrode - Beckman Model 39273 bright inlaid platinum electrode or equivalent.
b) Saturated Calomel Electrode (SCE) and associated salt bridge prepared in accordance with following procedures or equivalent.

1) Preparation of Saturated Calomel Electrode (SCE)

Equipment

a) 50 ml erlenmeyer flask
b) Double distilled mercury
c) Potassium chloride (KCl) certified ACS grade
d) Calomel (Hg_2Cl_2) - certified ACS grade
e) Mortar and pestle
f) 250 ml bottle

Procedure

Step 1: Thoroughly clean a 50 ml erlenmeyer flask with chromic acid cleaning solution, rinse with water and dry.
Step 2: Place sufficient mercury (double distilled) to form a layer 1 cm deep in the bottom of the flask.
Step 3: Prepare 200 ml of a saturated solution of KCl.
Step 4: To the above solution add about 2 grams of calomel (Hg_2Cl_2) and shake vigerously so as to obtain a saturated solution. It is best to prepare this solution a day in advance.

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Step 5: Place a few drops of merucury in a mortar and add approximately 7 grams of calomel, moisten with a few drops of the above calomel-saturated KCl solution and grind together until the mixture has a decided black or grayish black appearance.

Step 6: Wash the mixture several times by decantation with the clear supernatant liquid from the calomel saturated KCl solution.

Step 7: After the last decantation, transfer a sufficient amount of the mixture to form a 0.5 cm layer on top of the mercury.

Step 8: Add the remaining portion of the calomel-saturated KCl solution to the cell until it is about two-thirds full.

Step 9: Seal a 3-4 inch piece of platinum wire into a soft glass tube taking care that the seal does not leak. This is then inserted in a rubber stopper and adjusted so that the platinum wire is completely submerged below the surface of the mercury.

2) Preparation of Salt Bridge

Equipment

a) Agar - laboratory grade
b) Potassium chloride (KCl) - certified ACS grade.
c) 6-8 mm glass tubing
d) 200 ml beaker

Procedure

Step 1: Heat 3 grams of agar in 100 ml of water on a steam bath with stirring until the solution is clear.

Step 2: Add 30 grams of KCl and stir until dissolved

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Step 3: Suck the above solution into a 6-8 mm glass tube which has been bent to the desired configuration.

Step 4: If shrinkage of the gel occurs on setting, the glass is cut back with a file to expose it.

Step 5: When not in use, store the exposed end of the salt bridge in a IN KCl solution.

c) Two volt - One amp regulated power supply
d) High impedance voltmeter (10^{12} ohms or greater) equipped with recorder
e) Graphite rod electrode

4.4.3.15.2 Procedure

Step 1: Clean the platinum electrode in concentrated nitric acid and rinse well with distilled or deionized water.

Step 2: Immerse the electrode in a 1N solution of nitric acid and connect to the positive terminal of a constant potential power supply. Connect the negative terminal to a graphite rod also immersed in the acid solution.

Step 3: Set the voltage at 2.0 volts and activate the power supply for 1.0 minutes. Oxygen should be evolved freely at the electrode surface.

Step 4: Turn off the power supply and reverse the polarity of the electrodes. Reactivate for a period of 20 - 90 minutes at a voltage of 2.0 volts. Hydrogen must be evolved at the surface of the electrode throughout the entire cathodization process. If hydrogen evolution should cease, repeat the entire treatment process and if the problem is not corrected, the electrode shall be considered unsatisfactory and replaced.

Step 5: Remove the electrode from the acid solution, rinse well with distilled or deionized water and then turn off the power supply.

Step 6: Dry the electrode in a stream of cool air.

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Step 7: Rinse the tip of the calomel electrode's salt bridge with distilled or deionized water and dry in a stream of cool air.

Step 8: Connect the electrodes to a high impedance voltmeter (10^{12} ohms or greater) equipped with a recorder.

Step 9: Immerse the electrodes in a small beaker containing the test solution and activate the voltmeter and recorder. This measurement may be done under normal atmospheric conditions but the solution must be kept in a sealed container until the time of the measurement.

Step 10: Take the reading 400 seconds after the immersion of the electrodes. A certain amount of drift can be expected but the reading should be relatively stable after 400 seconds. If excessive drift is still occuring after this period ($>$ 0.3 mv/sec), the measurement shall be considered void.

Step 11: Repeat the above procedure until five values have been obtained and then average these values. A fresh sample shall be used for each measurement.

"The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment."

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